STAT 301 In-class Activity No. 10

Consider the "wage" dataset again. We now compare the following two (nested) linear regression models

$$\begin{aligned} \text{Model I:} \qquad & \text{wage} \;= \beta_0 + \beta_1 \text{ education } + e, \\ \text{Model II:} \qquad & \text{wage} \;= \beta_0 + \beta_1 \text{ education } + \beta_2 \text{ sex } + e. \end{aligned}$$

(a) Compare the (adjusted) coefficients of determination R^2 of the above two models. Which model is better?

(b) Is Model II significantly better than Model I? Answer the question by performing a F-test based on the R function "anova".